

Review 1

PROJ2999 Capstone Project Introduction

SYNAPSE-MAS: Shared Intelligent Memory Architecture for Multi-Agent LLM Systems

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Outline

- 1 Introduction
- 2 Literature Review
- 3 Summary of Literature Review
- 4 Research Gaps
- 5 Problem Statement
- 6 Research Objectives
- 7 Workplan and Proposed Timeline
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Multi-Agent Systems Need Memory

Why the problem matters

Specialized agents divide complex tasks and accumulate useful experience in parallel.

Agents can communicate during a task, but communication alone does not create durable, reusable knowledge.

Conventional stateless agents lose that experience across sessions.

Repeated rediscovery wastes computation, tokens and time.

Observed failure modes

Stale or contradictory memory may be reused.

Cross-agent knowledge transfer is weak and ad hoc.

Provenance, scope and consistency are difficult to govern.

The research need is a shared, selective and evolving memory layer.

Why Existing Approaches Fall Short

Stateless

No persistent experience,
no long-horizon continuity.

Retrieval-only / RAG

Useful for access to stored
context, but not inherently
designed for memory
admission, conflict
resolution or evolution.

Static shared memory

Shared state can become
stale, noisy or contradictory
without lifecycle control
and governance.

The missing capability

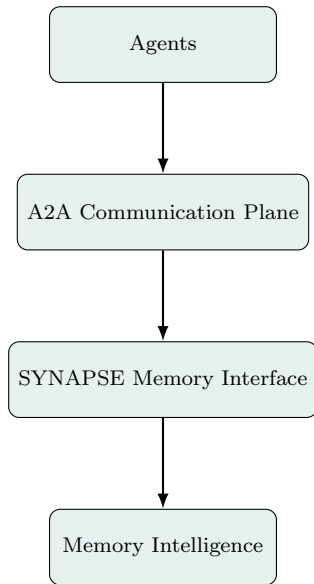
A memory architecture that can decide what to remember, what to reject, what to retrieve, and how to reconcile conflicting knowledge over time.

Research Question

Research Question

“Can a selective, persistent and evolving shared-memory architecture improve the long-horizon performance, knowledge transfer and consistency of multi-agent LLM systems compared with conventional stateless and retrieval-only approaches?”

SYNAPSE Architecture Concept



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Literature Review: Overview

Focus of the review

This review examines how multi-agent LLM systems communicate, share knowledge and maintain useful memory across tasks and sessions.

Themes across the literature

Memory formation: deciding what experience is worth retaining.

Shared memory: enabling agents to access common knowledge safely.

Architecture: organizing memory hierarchy, access and consistency.

Governance: controlling scope, provenance, ownership and updates.

Six-paper review path

Selective persistent memory

Governed shared memory

Computer-architecture perspective

Mesh memory protocol

Adaptive memory lifecycle

Change-intent and write governance

Literature Review (Paper 1)

Title of the paper [?]

Shared Selective Persistent Memory for Agentic LLM Systems

Sanjana Pedada, Aditya Dhavala, Neelraj Patil (2026).

Problem: preserve reusable context across multi-turn agent sessions without retaining irrelevant reasoning history.

Main contributions

Selective retention of task specifications, data schemas, tool configurations and output constraints.

Shared persistent workspaces with role-based access.

Separates reusable context from session-specific reasoning traces.

Introduces zero-token data refresh for recurring updates.

Methodology

Selective shared persistent memory

Identify reusable context categories.

Discard session-specific traces and persist selected knowledge.

Reuse stored context across sessions/users.

Evaluate task completion and refresh efficiency.

Tools and Technology used

LLM-based agentic code-generation system; persistent memory/workspace; git-versioned artifacts; heterogeneous data sources; role-based access control.

Summary of Results

96% task completion with memory vs. **79%** without memory and **71%** with full history in the reported enterprise scenarios. Zero-token refresh reduced task time by **14×**; summary-driven generation reduced token cost by **97×**.

Major Findings

Memory quality matters more than memory quantity.

Full-history persistence can degrade performance because of stale traces.

Selective memory supports collaborative reuse and efficiency.



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Literature Review (Paper 2)

Title of the paper [?]

Governed Shared Memory for Multi-Agent LLM Systems

Yanki Margalit, Nurit Cohen-Inger, Erni Avram, Ran Taig, Oded Margalit (2026).

Problem: shared multi-agent memory can suffer from unauthorized leakage, stale propagation, contradictions and loss of provenance.

Main contributions

- Formalizes the fleet-memory problem.
- Defines scoped retrieval and temporal supersession.
- Adds provenance tracking and policy-governed propagation.
- Evaluates governance through a reproducible harness.

Methodology

Governed shared-memory service + evaluation harness

- Define four governance failure modes.
- Apply scope and propagation policies.
- Track writer identity and derivation lineage.
- Test governance behaviour under live service conditions.

Tools and Technology used

MemClaw multi-tenant memory service; ArgusFleet evaluation harness; scoped retrieval; provenance and policy controls.

Summary of Results

Reported 100% reconstruction of depth-four provenance chains with correct writer identity and zero cross-fleet leakage under the tested conditions. The study also exposed scope-enforcement and pipeline-ordering issues.

Major Findings

- Long-context retrieval alone is insufficient for governed shared memory.
- Governance must be explicit at the systems level.
- Pipeline ordering can affect contradiction handling.

Literature Review (Paper 3)

Title of the paper [?]

Multi-Agent Memory from a Computer Architecture Perspective: Visions and Challenges Ahead

Zhongming Yu, Naicheng Yu, Hejia Zhang, Wentao Ni, Mingrui Yin, Jiaying Yang, Yujie Zhao, Jishen Zhao (2026).

Problem: multi-agent memory is becoming a systems/architecture bottleneck involving hierarchy, access and consistency.

Main contributions

Frames multi-agent memory as a computer-architecture problem.

Distinguishes shared and distributed memory paradigms.

Proposes a three-layer I/O-cache-memory hierarchy.

Identifies cache-sharing and structured access control

Methodology

Architectural / position framework

Compare shared vs. distributed memory.

Map I/O, cache and memory layers to agent systems.

Examine access protocols and consistency requirements.

Identify open research directions.

Tools and Technology used

Computer-architecture abstractions; memory hierarchy; access protocols; consistency mechanisms.

The paper is a position paper rather than an empirical model benchmark.

Summary of Results

The paper argues that **multi-agent memory consistency** is a central open challenge and that hierarchical memory plus explicit protocols are needed for reliable, scalable systems.

Major Findings

Memory should be treated as an architectural subsystem.

Consistency is not solved merely by providing shared storage.

Protocol design and structured access remain open problems.



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Literature Review (Paper 4)

Title of the paper [?]

Mesh Memory Protocol: Semantic Infrastructure for Multi-Agent LLM Systems

Hongwei Xu (2026).

Problem: agents collaborating across sessions need a semantic protocol for deciding what peer knowledge to accept, tracing its origin and preserving useful knowledge after restart.

Main contributions

- Defines field-level rather than whole-message admission.
- Provides inter-agent lineage for claims.
- Stores the receiver's role-evaluated understanding rather than raw peer signals.
- Specifies a semantic infrastructure layer for cross-session collaboration.

Methodology

Mesh Memory Protocol (MMP)

- CAT7 represents Cognitive Memory Blocks.
- SVAf evaluates fields against role-indexed anchors.
- Content-hash parents/ancestors provide lineage.
- Remix stores the receiver's evaluated understanding.

Tools and Technology used

Cognitive Memory Blocks (CMB); CAT7 schema; SVAf admission; content-hash lineage; receiver-side remix; agent identity and persistent memory.

Summary of Results

The paper reports three reference deployments in which autonomous agents collaborate across sessions using the specified semantic memory protocol.

Major Findings

- Shared memory should not blindly copy peer messages.
- Admission, provenance and receiver-side interpretation are distinct requirements.
- This directly supports SYNAPSE's selective-admission and lineage design.



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Literature Review (Paper 5)

Title of the paper [1]

AMA: Adaptive Memory via Multi-Agent Collaboration

arXiv:2601.20352 (2026).

Problem: multi-agent systems need memory that can adapt to task context and uncertainty instead of relying on a fixed memory schema.

Main contributions

Introduces explicit memory lifecycle stages across formation and update.

Uses varying retrieval granularity and query routing.

Incorporates judgement, retry and conflict detection.

Supports refresh and targeted memory update when evidence is insufficient.

Methodology

Adaptive memory lifecycle

Extract reusable experience from agent actions.

Route retrieval based on task query and memory granularity.

Judge reliability and sufficiency of evidence.

Refresh or revise memory when conflicts or gaps emerge.

Tools and Technology used

Multi-agent collaboration patterns; adaptive retrieval; lifecycle-based memory control; evidence-aware update logic.

Summary of Results

The paper highlights adaptive memory as a dynamic lifecycle, where agents decide when to retrieve, when to retry and when to revise memory based on evidence quality.

Major Findings

Memory must adapt to task context and uncertainty.

Retrieval should not be fixed at one granularity.

Judgement and refresh are necessary for durable memory quality.

SYNAPSE Adapts

SYNAPSE adopts the lifecycle idea as formation → selective admission → judgement → conflict resolution →

Literature Review (Paper 6)

Title of the paper [2]

Claim Plane: Enforceable Change Intents and Dynamic Scope

arXiv:2607.21909 (2026).

Problem: concurrent agents need a disciplined mechanism for deciding whether a write or change should be admitted under ambiguity or conflict.

Main contributions

- Defines pre-write admission and intent tracking.

- Emphasizes fail-closed behavior under ambiguity.

- Supports change scope and version-aware enforcement.

- Links provenance to write control decisions.

Methodology

Deterministic write admission

- Define the intended change before write execution.

- Validate scope and version constraints.

- Refuse ambiguous or unsafe updates.

- Preserve lineage and provenance metadata.

Tools and Technology used

Deterministic scope enforcement, change-intent governance, version tracking, provenance-aware update control, fail-closed decision logic.

Summary of Results

The paper presents a strong engineering principle: when intent is uncertain, the correct system behavior is conservative rejection rather than unsafe propagation.

Major Findings

- Intent-aware admission is critical in concurrent settings.
- Provenance and versioning are not optional extras.

- Fail-closed behavior improves reliability under ambiguity.

SYNAPSE Adapts

SYNAPSE uses the same principle for memory-write governance, conflict admission, lineage tracking and controlled memory evolution.

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Summary of Literature Review

Ref.	Overview of Paper	Model / Approach	Novelty / Contribution	Limitations / Scope
[3]	Selective persistent memory	structured memory + admission	reusable task specifications and memory quality	enterprise/workspace context; no shared multi-agent governance
[4]	Governed shared memory	provenance + temporal validity + scope control	policy-governed shared state	governance-centric, not full lifecycle intelligence
[5]	Architecture framing	memory hierarchy and consistency view	shared vs distributed memory analysis	conceptual/position framing
[6]	Mesh memory protocol	receiver-side semantic admission + lineage	semantic infrastructure for exchange	protocol-level scope, not full memory intelligence
[1]	Adaptive memory	lifecycle roles + targeted refresh	judgement, conflict, granularity	different architecture/context
[2]	Change-intent governance	deterministic write admission + scope enforcement	provenance + fail-closed control	coding-agent setting; not a general memory architecture



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Research Gaps

Observations from Literature

The literature shows value in selective memory, provenance, retrieval relevance, and layered multi-agent architecture, but these ideas are typically addressed in isolation rather than as one integrated memory lifecycle.

Identified Gaps

Selective admission and memory quality. Existing work demonstrates the value of selective memory, but SYNAPSE studies this in a shared multi-agent setting with additional judgement and lifecycle control.

Governed shared memory. Prior work addresses provenance, scope and temporal validity, but SYNAPSE integrates governance across formation, retrieval, update and evolution.

Semantic conflict resolution. Storage consistency alone does not guarantee semantic consistency; SYNAPSE treats judgement and conflict resolution as memory-intelligence operations.

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Problem Statement

Problem Statement

To design and evaluate a shared intelligent memory architecture for multi-agent LLM systems that enables agents to selectively store, retrieve, share, consolidate and evolve knowledge and experiences across interactions, while maintaining relevance, consistency, provenance and efficient memory usage.

Research Direction

SYNAPSE-MAS studies a shared memory architecture in which memory is a governed coordination substrate rather than a passive vector store or an implicit conversation transcript.



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Project Objectives

Objectives

Design and implement a shared memory architecture supporting persistent episodic and semantic memory with contextual retrieval.

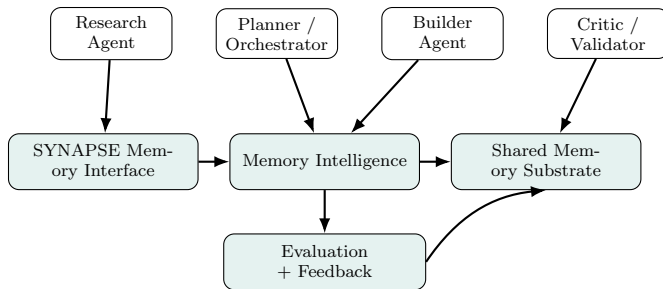
Develop selective memory formation and evolution mechanisms for relevance-based admission, redundancy reduction, conflict resolution, consolidation and updating/evolution.

Evaluate SYNAPSE-MAS against the stateless baseline, the retrieval-only / RAG baseline and the static/shared-memory baseline using task performance, retrieval quality, memory efficiency, cross-agent knowledge transfer and consistency.

Evaluation metrics

Measured outcomes include task performance, retrieval quality, memory efficiency, cross-agent knowledge transfer, consistency, token usage, latency and storage growth.

Proposed System Direction – SYNAPSE-MAS



Key idea: A2A supports communication, while SYNAPSE governs the durable memory plane for retrieval, validation, conflict handling and memory evolution.



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Work Breakdown and Task Allocation

Task Area	Primary Owner	Key Responsibilities
Problem framing and literature synthesis	Krishna	Identify research question, review core papers, derive gaps and structure the Review-I presentation
Architecture and agent interfaces	Sanjay	Define agent layers, A2A communication, memory interface and orchestration boundaries
Storage, indexing and retrieval	Ram	Persistent storage design, structured and vector memory, retrieval APIs, metadata handling
Memory intelligence and testing	Aman	Selective admission, conflict handling, consolidation, evaluation design and ablation tests
End-to-end integration and analysis	All	Prototype integration, validation, reporting and Review-II preparation



Proposed Timeline and Milestones

Milestone	Status	Window	Owner	Focus
M1: Problem formulation and literature review	Completed	Sep 2026	Krishna	Research question, problem statement, literature synthesis
M2: Architecture and requirements baseline	In Progress	Oct 2026	Sanjay + Ram	Memory interface, component boundaries, governance requirements
M3: Shared-memory prototype	Planned	Oct-Nov 2026	Ram + Sanjay	Storage, retrieval, metadata, provenance
M4: Memory intelligence and conflict mechanisms	Planned	Nov 2026	Aman	Admission, judgment, consolidation, evolution
M5: Baselines and experimental evaluation	Planned	Nov-Dec 2026	All	Task performance, memory efficiency, transfer and consistency
M6: Review-II preparation	Planned	Dec 2026	All	Analysis, reporting and presentation refinement



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Action Plan

Freeze the architecture boundary between A2A communication and SYNAPSE memory operations.

Implement the memory interface with storage, retrieval and provenance metadata.

Implement persistent episodic and semantic memory with contextual retrieval using a minimal prototype pipeline.

Add memory-intelligence services for selective admission, redundancy reduction, judgement and conflict resolution.

Implement consolidation and evolution routines that update memory lineage and maintain semantic consistency.

Compare SYNAPSE-MAS against stateless, retrieval-only and static shared-memory baselines.

Analyze task performance, retrieval quality, memory efficiency, transfer and consistency to prepare Review-II.

References I



Anonymous authors.

Ama: Adaptive memory via multi-agent collaboration.
arXiv:2601.20352, 2026.



Anonymous authors.

Claim plane: Enforceable change intents and dynamic scope.
arXiv preprint arXiv:2607.21909, 2026.



Sanjana Pedada, Aditya Dhavala, and Neelraj Patil.

Shared selective persistent memory for agentic llm systems.
arXiv preprint arXiv:2607.09493, 2026.



Yanki Margalit, Nurit Cohen-Inger, Erni Avram, Ran Taig, and Oded Margalit.

Governed shared memory for multi-agent llm systems.
arXiv preprint arXiv:2606.24535, 2026.



Anonymous authors.

Multi-agent memory from a computer architecture perspective: Visions and challenges ahead.
arXiv preprint arXiv:2603.10062, 2026.



Hongwei Xu.

Mesh memory protocol: Semantic infrastructure for multi-agent llm systems.
arXiv preprint arXiv:2604.19540, 2026.



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Disclosure Statements

I will maintain academic honesty by doing my own work and avoiding plagiarism, cheating, unauthorized collaboration, falsifying information, and misuse of AI-generated content.

I will properly acknowledge and cite all sources used, including papers, websites, software and AI tools.

I will use institutional resources responsibly and only for authorized academic or research activities.

I understand that I remain responsible for the correctness, originality and decisions in the submitted work.



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